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The Identification Capacity Region of the Broadcast Channel

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ABSTRACT

In this work we characterize the identification capacity region of a general broadcast channel. The direct part is based on code construction which makes use of the common randomness inherent to any transmission coding scheme with positive common message rate. The converse part makes use of a distribution packing lemma, applied to each of the message sets separately.

Index terms — Broadcast channel, common experiment, degraded message sets, distribution packing, identification via channels.